

Restaurant Recommendation System Using Content-Based Filtering

1. Objective

The objective of this project is to develop a restaurant recommendation system that suggests restaurants based on user preferences such as cuisine type and price range using content-based filtering techniques.

2. Dataset Information

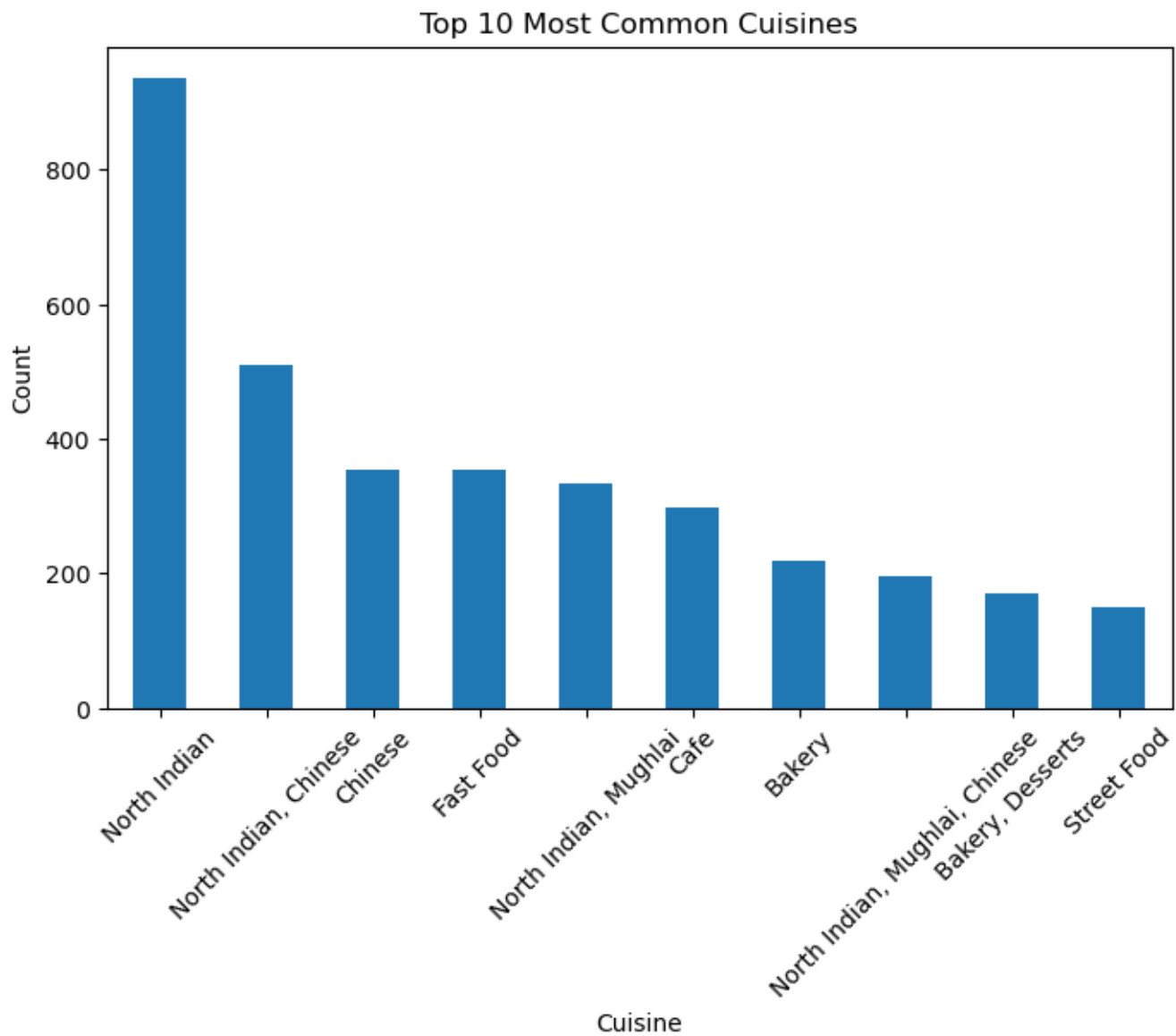
- Total Records: 9551
 - Records After Cleaning: 9542
 - Missing Values Removed: 9
 - Features Used:
 - Restaurant Name
 - Cuisines
 - Price Range
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3. Data Preprocessing

- Removed missing values from the Cuisines column.
 - Selected Cuisine and Price Range as recommendation criteria.
 - Combined recommendation features into a single text feature.
 - Prepared data for TF-IDF vectorization.
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4. Exploratory Data Analysis

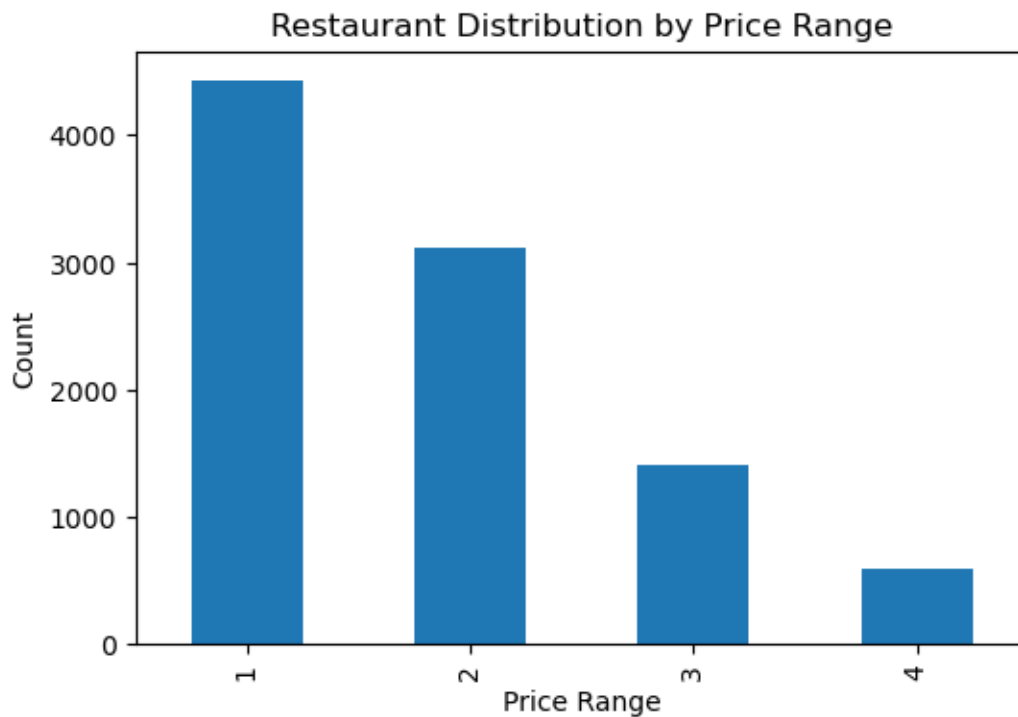
Top 10 Most Common Cuisines



Observation:

- North Indian cuisine appears most frequently in the dataset.
- Several cuisine categories are highly represented.

Price Range Distribution



Observation:

- Most restaurants belong to lower and medium price categories.
- High-price restaurants are comparatively fewer.

5. Recommendation System Development

TF-IDF Vectorization

TF-IDF (Term Frequency–Inverse Document Frequency) was applied to transform textual recommendation features into numerical vectors.

Output Shape:

(9542, 150)

Cosine Similarity

Cosine Similarity was used to measure similarity between restaurants based on cuisine and price range features.

Output Shape:

(9542, 9542)

6. Recommendation Results

Example 1: Cafe Arabelle

Recommended Restaurant	Cuisines	Price Range
Locavore	Filipino	3
Guevarra's	Filipino	3
Balay Dako	Filipino	3
Silantro Fil-Mex	Filipino, Mexican	3
Silantro Fil-Mex	Filipino, Mexican	3

Observation:

The system recommended restaurants serving Filipino cuisine with similar price ranges.

Example 2: Le Petit Souffle

Recommended Restaurant	Cuisines	Price Range
Tokyo Mon Amour	Japanese, French	4
Paris 6 Classique	French	4
Django	French	3
Jaan	French	4
Rhubarb Le Restaurant	French	4

Observation:

The system recommended restaurants offering French cuisine with similar characteristics.

Example 3: Mad Mark's Creamery & Good Eats

Recommended Restaurant	Cuisines	Price Range
Cryo Lab	Desserts, Ice Cream	2
Cream Stone	Desserts, Ice Cream	1
Giani's	Ice Cream, Desserts	1
Gelato Vinto	Desserts, Ice Cream	1
Giani's	Ice Cream, Desserts	1

Observation:

The system recommended restaurants specializing in Desserts and Ice Cream with comparable pricing.

7. Evaluation

The recommendation quality was evaluated manually by comparing recommended restaurants with the selected restaurant.

Results indicate that the recommended restaurants generally share:

- Similar cuisine categories

- Similar price ranges
- Comparable restaurant characteristics

This demonstrates the effectiveness of content-based filtering.

8. Challenges

- Recommendations rely heavily on cuisine information.
 - Restaurants with unique cuisine combinations may have fewer similar matches.
 - User ratings and customer reviews were not included in recommendation generation.
 - Recommendation quality depends on the completeness of restaurant feature information.
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9. Conclusion

The Restaurant Recommendation System successfully generated relevant restaurant recommendations using content-based filtering techniques. By applying TF-IDF Vectorization and Cosine Similarity, the system was able to identify restaurants with similar cuisines and price ranges, producing meaningful and useful recommendations for users.